

Homework 1

p 10/ex 11

$$\begin{cases} X_2 + 4X_3 = -5 \\ X_1 + 3X_2 + 5X_3 = -2 \\ 3X_1 + 7X_2 + 7X_3 = 6 \end{cases}$$

$$\begin{bmatrix} 0 & 1 & 4 & -5 \\ 1 & 3 & 5 & -2 \\ 3 & 7 & 7 & 6 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 4 & -5 \\ 3 & 7 & 7 & 6 \end{bmatrix} \begin{matrix} R_1 - R_2 \cdot 3 \\ R_3 - R_1 \cdot 3 \end{matrix}$$

$$\begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 4 & -5 \\ 0 & -2 & -8 & 12 \end{bmatrix} \begin{matrix} R_1 - R_2 \cdot 3 \\ R_3 / 2 \end{matrix} \Rightarrow \begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & 1 & 4 & -5 \\ 0 & -1 & -4 & 6 \end{bmatrix} \begin{matrix} R_1 - R_2 \cdot 3 \\ R_3 + R_2 \end{matrix}$$

$$\begin{bmatrix} 1 & 0 & -7 & 13 \\ 0 & 1 & 4 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow 0 \neq 1 \Rightarrow \text{no real solution} \\ \text{inconsistent}$$

p10/ex n 13

$$\begin{cases} x_1 - 3x_3 = 8 \\ 2x_1 + 2x_2 + 8x_3 = 7 \\ x_2 + 5x_3 = -2 \end{cases}$$

$$\begin{bmatrix} 1 & 0 & -3 & | & 8 \\ 2 & 2 & 8 & | & 7 \\ 0 & -1 & 5 & | & -2 \end{bmatrix}$$

$R_2 - 2 \cdot R_1 \Rightarrow$

$$\begin{bmatrix} 1 & 0 & -3 & | & 8 \\ 0 & 2 & 14 & | & -9 \\ 0 & -1 & 5 & | & -2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -3 & | & 8 \\ 0 & 1 & 5 & | & -2 \\ 0 & 2 & 14 & | & -9 \end{bmatrix}$$

$\Rightarrow$

$$\begin{bmatrix} 1 & 0 & -3 & | & 8 \\ 0 & 1 & 5 & | & -2 \\ 0 & 0 & 4 & | & -5 \end{bmatrix}$$

$R_2 \leftrightarrow R_3$

$$\Rightarrow \begin{bmatrix} 1 & 0 & -3 & | & 8 \\ 0 & 1 & 0 & | & 3 \\ 0 & 0 & 1 & | & -1 \end{bmatrix}$$

$R_1 + R_3 \cdot 3$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 5 \\ 0 & 1 & 0 & | & 3 \\ 0 & 0 & 1 & | & -1 \end{bmatrix}$$

$$\begin{cases} x_1 = 5 \\ x_2 = 3 \\ x_3 = -1 \end{cases}$$

p10/ex 24

$\begin{cases} x_1 + 3x_2 = f \\ cx_1 + dx_2 = g \end{cases}$ if these equations are consistent for all possible answers it means they are the same equation:

$$f = g \Rightarrow x_1 + 3x_2 = cx_1 + dx_2$$

Hence, $c = 1$; $d = 3$.

p22/ex N 13

$$\begin{bmatrix} 1 & -3 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 8 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{+R_2 \cdot 3 + R_3}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 0 & -3 & 5 \\ 0 & 1 & 0 & 0 & -4 & 1 \\ 0 & 0 & 0 & 1 & 8 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} x_1 - 3x_5 = 5 \\ x_2 - 4x_5 = 1 \\ x_4 + 8x_5 = 4 \end{cases}$$

$$\begin{cases} x_1 = 5 + 3x_5 \\ x_2 = 1 + 4x_5 \\ x_4 = 4 - 8x_5 \end{cases} \quad \begin{array}{l} x_1, x_2, x_4 = \text{basic var} \\ x_3, x_5 = \text{free variable} \end{array}$$

p40/ex N 5

$$\begin{bmatrix} 5 & 1 & -8 & 4 \\ -2 & -7 & 3 & -5 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \\ 3 \\ -2 \end{bmatrix} = \begin{bmatrix} -8 \\ 16 \end{bmatrix} \Rightarrow$$

$$\Rightarrow 5 \begin{bmatrix} 5 \\ -2 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ -7 \end{bmatrix} + 3 \begin{bmatrix} -8 \\ 3 \end{bmatrix} - 2 \begin{bmatrix} 4 \\ -5 \end{bmatrix} = \begin{bmatrix} -8 \\ 16 \end{bmatrix}$$

p40 / ex N 7

$$x_1 \begin{bmatrix} 4 \\ -1 \\ 7 \\ -4 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ 3 \\ -5 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ -8 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix} \Rightarrow$$

$$\Rightarrow \begin{bmatrix} 4 & -5 & 7 \\ -1 & 3 & -8 \\ 7 & -5 & 0 \\ -4 & 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}$$

p10 / ex N 15

$$\begin{bmatrix} 1 & 0 & 3 & 0 & | & 2 \\ 0 & 1 & 0 & -3 & | & 3 \\ 0 & -2 & 3 & 2 & | & 1 \\ 3 & 0 & 0 & 7 & | & -5 \end{bmatrix} \begin{array}{l} \\ \\ +R_2:2 \\ -R_2:3 \end{array} \Rightarrow \begin{bmatrix} 1 & 0 & 3 & 0 & | & 2 \\ 0 & 1 & 0 & -3 & | & 3 \\ 0 & 0 & 3 & -4 & | & 7 \\ 0 & 0 & -3 & 7 & | & -11 \end{bmatrix} \begin{array}{l} -R_3 \\ \\ \\ +3R_3 \end{array} \Rightarrow$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & 4 & | & -5 \\ 0 & 1 & 0 & -3 & | & 3 \\ 0 & 0 & 3 & -4 & | & 7 \\ 0 & 0 & -3 & 7 & | & 10 \end{bmatrix}$$

System is consistent